

EXPERIMENT 5

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Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment -5

Implementation of Dynamic Routing using RIP (Routing Information Protocol) **Aim**

To implement Dynamic Routing using RIP (Routing Information Protocol) in Cisco Packet Tracer and understand how routers automatically learn and update routing information between different networks.

Objective

- To understand the concept of dynamic routing.
- To study the working of Routing Information Protocol (RIP).
- To configure RIP on routers using Cisco IOS commands.
- To understand how routers automatically exchange routing information.
- To verify communication between different networks using dynamic routing.
- To understand hop count as the routing metric used by RIP.

Software Requirements

- Cisco Packet Tracer

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Hardware (Virtual)

- Routers
- Switches
- PCs
- Copper Straight-Through Cables

Theory

Routing is the process of finding the best path for forwarding data packets from a source network to a destination network. Routers are required when communicating between different networks.

Dynamic routing is a routing technique in which routers automatically learn and update routing information. Routers exchange routing information with their neighboring routers, reducing the need for manually configured routes.

RIP stands for Routing Information Protocol. It is a distance-vector routing protocol that automatically exchanges routing information between routers. RIP uses hop count as its routing metric. The maximum reachable hop count is 15, while a hop count of 16 represents an unreachable network.

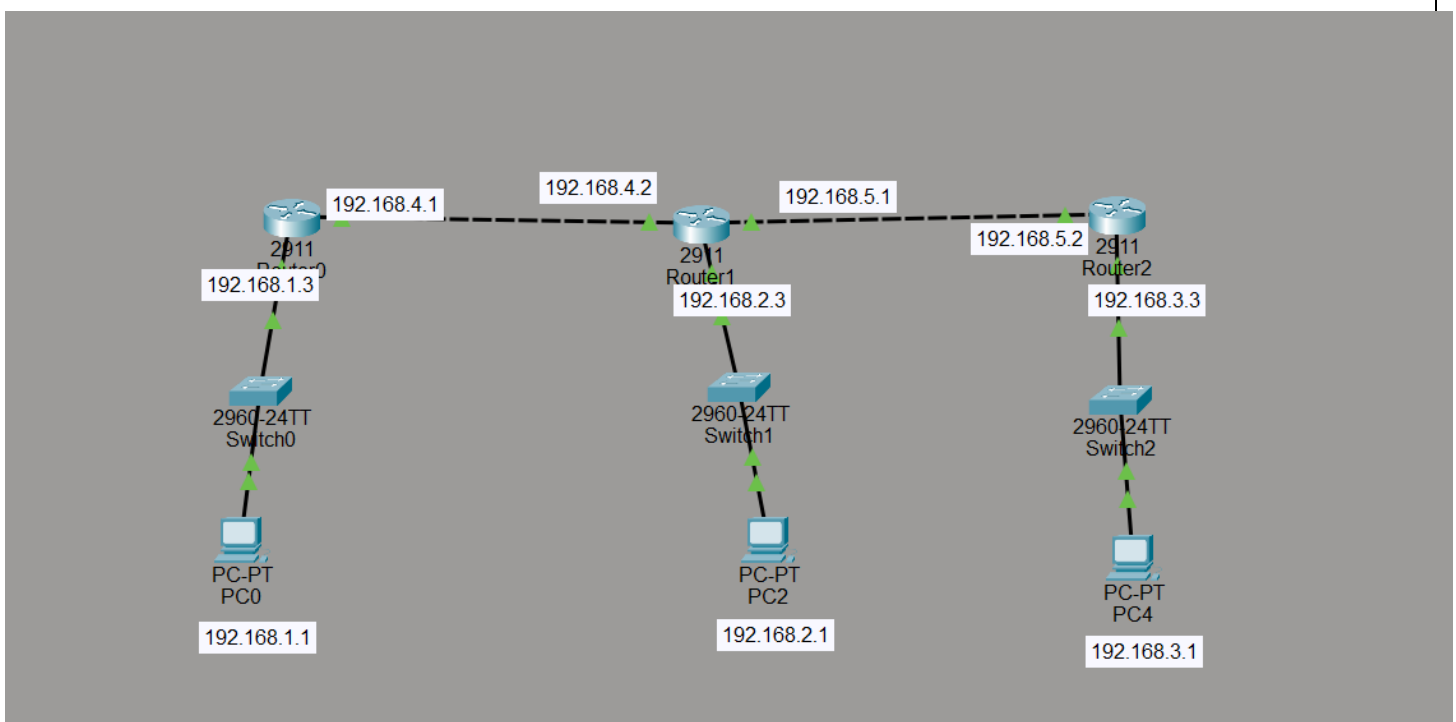
Practical / Experiment Steps

1. Open Cisco Packet Tracer.
2. Place the required routers, switches, and PCs in the workspace.
3. Connect the PCs to the switches.
4. Connect the switches to the routers.
5. Connect the routers with appropriate cables.
6. Assign IP addresses to the PCs and router interfaces.

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7. Configure the router interfaces with their respective IP addresses.
8. Enable RIP on the routers.
9. Add the connected networks to RIP configuration.
10. Verify the routing table of each router.
11. Use the Ping command to test communication between different networks.

Observe that the routers automatically learn the routes through RIP. **RIP**



Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand the concept of dynamic routing** and its role in enabling communication between multiple networks by automatically learning and updating routing information.

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2. **Develop practical skills in configuring network devices** by assigning IP addresses, subnet masks, default gateways, and router interfaces in Cisco Packet Tracer.